SWA-MPPI: Socially-Weighted Alignment with Model Predictive Path Integral Control for Cross-Cultural Moral Alignment of Large Language Models

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Abstract

Large language models inherit the moral intuitions of their predominantly Western training data, yet serve culturally diverse populations worldwide. No existing method can adapt an LLM's moral reasoning to a target culture *at inference time* without labelled preference data or weight modification. We introduce **SWA-MPPI**, the first training-free framework for cross-cultural moral alignment that operates entirely in logit space. SWA-MPPI constructs an ensemble of culturally grounded persona agents from World Values Survey data, detects inter-persona disagreement as a signal of cultural contestation, and resolves it via a Prospect-Theory-weighted Model Predictive Path Integral (MPPI) update—a principled stochastic optimisation that balances private cultural utility with collective welfare. On the Multilingual Trolley Problem benchmark across 15 countries, 12 native languages, and 6 moral dimensions, SWA-MPPI reduces Jensen-Shannon Distance to human moral preferences by **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B), outperforming all prompting and activation-steering baselines by $\geq$10% while operating on frozen weights with a single additional batched forward pass. Notably, the largest gains occur for Anglosphere countries (+72–75%), revealing that RLHF-induced biases actively worsen alignment even for the best-represented populations. The framework is self-limiting: when the model is already well-calibrated, inter-persona consensus is high and no correction is applied. Code available at [repository].

1 Introduction

When a self-driving car must choose between sparing a group of elderly men or a group of young women, there is no culturally universal answer. The Moral Machine experiment [Awad et al., 2018]—40 million judgments from 233 countries—revealed that such preferences vary profoundly across cultures: East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to assist with such decisions encode a single, culturally monolithic moral stance inherited from their predominantly Western training data [Henrich et al., 2010].

The multilingual successor benchmark MultiTP [Jin et al., 2025] operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages, providing a rigorous test bed for cross-cultural alignment. The verdict is stark: Jin et al. [2025] found that none of 19 contemporary LLMs achieved broadly satisfactory alignment—most exhibit WEIRD-biased reasoning that departs markedly from non-WEIRD human preferences.

How should this gap be closed? Existing approaches each sacrifice critical properties. *Fine-tuning* [Bai et al., 2022, Ouyang et al., 2022] demands per-culture labelled data and modifies weights irreversibly.

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Prompting-based steering relies on ad-hoc instructions with no empirical grounding [Jin et al., 2025]. *Activation steering* [Arditi et al., 2025, Zou et al., 2023] requires white-box access and per-model engineering. *Reward-guided decoding* [Khanov et al., 2024, Mudgal et al., 2024] needs trained reward models unavailable for cross-cultural settings. *Culture-aware adapters* [Li et al., 2024, Yao et al., 2024] scale poorly across 100+ countries. None of these methods can adapt an LLM's moral reasoning to a target culture *at inference time* without labelled data or weight modification.

We propose **SWA-MPPI** (Socially-Weighted Alignment with Model Predictive Path Integral control), built on a simple insight: *disagreement is signal*. Moral preferences within any society are not monolithic—age cohorts and social strata exhibit systematically different priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] provides empirically calibrated data on these within-country differences across 80+ nations. When culturally grounded persona agents derived from WVS data *disagree* on a moral decision, that disagreement reveals the decision is culturally contested—and can be exploited through stochastic optimal control [Williams et al., 2018] to produce a more culturally representative output.

Concretely, SWA-MPPI constructs $N = 4$ WVS-grounded personas per country (three age cohorts plus a utilitarian agent), evaluates all $N + 1$ agents in a single batched forward pass with native-language prompting across 12 languages, and extracts decision logits. If inter-agent variance exceeds a calibrated threshold τ, it samples $K = 128$ perturbations and scores each via a Prospect Theory value function [Kahneman and Tversky, 1979] that captures the asymmetric gain-loss sensitivity characteristic of human moral judgment. The importance-weighted MPPI update yields a minimal-KL correction; a positional debiasing step cancels recency effects.

Contributions.

- We introduce SWA-MPPI, the first application of MPPI in logit space for cross-cultural value alignment: a training-free framework combining WVS-grounded personas, adaptive conflict detection, and Prospect-Theory-weighted optimisation (§3).
- We demonstrate mean JSD reductions of **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B) across 15 countries, with gains reaching $+75\%$ for Anglosphere pairs, outperforming all baselines by $\geq 10\%$ (§5).
- We conduct systematic ablations revealing a clear component hierarchy: the utilitarian persona is most critical (-39.1% relative Pearson without it), followed by MPPI ($+15.9\%$ JSD), debiasing ($+12.0\%$), and Prospect Theory ($+4.5\%$) (§6).

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs—we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-MPPI targets binary moral decisions specifically.

Cross-cultural LLM evaluation. LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-MPPI's calibration objective.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models—unavailable for cross-cultural settings (Appendix G). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. SWA-MPPI is the first *training-free* method addressing multi-cultural calibration. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-MPPI operationalises concretely.

MPPI and Prospect Theory. MPPI [Williams et al., 2018] is a sampling-based stochastic optimal control algorithm originally developed for robotics; Huang et al. [2024] adapted it for token-level LLM steering. We are the first to apply MPPI in *logit space* for moral alignment, combining it with Prospect Theory [Kahneman and Tversky, 1979] to capture asymmetric gain-loss sensitivity in human moral judgment.

3 Method: SWA-MPPI

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `LEFT` (ℓ) and `RIGHT` (r). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_\ell,\ z_r]$; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_r - z_\ell$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention. For the five binary dimensions (Species, Gender, Age, Fitness, Social Value), this reduces to the empirical mean of p_{spare}. For the Utilitarianism dimension, where the predictor is the *count difference* $|n_{\text{pref}} - n_{\text{non-pref}}|$ between groups, we fit $p_{\text{spare}} \sim a + b \cdot n_{\text{diff}}$ via OLS and evaluate at the mean observed n_{diff} (Appendix A).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$ under Jensen-Shannon Divergence.

3.2 Cultural Persona Construction

For each country c, SWA-MPPI constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. We extract 8 value dimensions from specific WVS items: *gender egalitarianism* (Q58P, Q59P, Q60P), *religious importance* (Q6P), *interpersonal trust* (Q43P), *moral permissiveness* (Q50, Q52P, Q54P), *work centrality* (Q5P), *family importance* (Q1P), *personal autonomy* (Q39P), and *meritocratic orientation* (Q40P). For each country, respondents are segmented into three age cohorts using birth year and survey year: *young* (18–35), *middle-aged* (36–55), and *older* (>55). Dimension means are computed per cohort and converted into natural-language descriptors (e.g., a mean religiosity ratio >0.85 yields "deeply religious"). A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a natural-language system prompt:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} (Table 1) scale logits before computing decision gaps:

$$\delta_i = \frac{z_{i,r} - z_{i,\ell}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N} \sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*—a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit.

3.4 Adaptive Conflict Detection

When personas agree, the consensus $\bar{\delta}$ suffices. When they disagree substantially, a controlled correction is warranted. SWA-MPPI uses a per-country threshold $\tau^{(c)}$ calibrated on $n_{\text{cal}} = 50$ unlabelled scenarios to target a 35% MPPI trigger rate:

$$\tau^{(c)} = \text{Percentile}_{65}\Big(\big\{\text{Var}(\delta_1^{(s)}, \ldots, \delta_N^{(s)})\big\}_{s=1}^{n_{\text{cal}}}\Big). \tag{5}$$

MPPI fires iff $\text{Var}(\delta_1, \ldots, \delta_N) \geq \tau^{(c)}$. Calibration uses only variance statistics and never observes human AMCEs (Appendix E).

3.5 Prospect-Theory-Weighted MPPI Optimisation

When triggered, SWA-MPPI draws $K = 128$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ ($\sigma = 0.3$) and evaluates a collective welfare score for each $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

Per-agent cooperative utility.

$$U_i(\epsilon_k) = (1 - \lambda_{\text{coop}})\, v(r_i\, \tilde{\delta}_k) \;+\; \lambda_{\text{coop}}\, v\Big(\bar{r}_{-i}\, \tilde{\delta}_k\Big), \tag{6}$$

where $\lambda_{\text{coop}} = 0.7$ and $v(\cdot)$ is the Prospect Theory value function with $\alpha = \beta = 0.88$, $\kappa = 2.25$ [Kahneman and Tversky, 1979]:

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\, |x|^{\beta} & x < 0. \end{cases} \tag{7}$$

Collective utility with KL regularisation.

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N} \sum_{i=1}^{N} U_i(\epsilon_k) \;-\; \alpha_{\text{KL}}\, \frac{\epsilon_k^2}{2\sigma^2}, \tag{8}$$

where $\alpha_{\text{KL}} = 0.05$. The KL term derives from the free-energy objective of MPPI, preventing overcorrection in a theoretically grounded way (Appendix C).

MPPI update.

$$w_k = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\beta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\beta\big)}, \qquad \delta^* = \sum_{k=1}^{K} w_k\, \epsilon_k, \tag{9}$$

where $\beta = T_{\text{dec}} = 0.5$ is the MPPI inverse temperature that governs the sharpness of importance weights. We set β equal to the decision temperature by design: this ensures that the MPPI optimisation operates on the same scale as the final sigmoid decision, so that a perturbation ϵ_k that shifts δ_{opt} by one unit produces a proportionate change in p_{spare}. Lower β would concentrate weights on fewer high-utility samples (sharper selection); higher β would spread weights more uniformly (softer averaging). At $\beta = 0.5$, the effective sample size $K_{\text{eff}} = 1/\sum_k w_k^2$ is typically 15–40 (out of $K = 128$), indicating moderate concentration that balances exploration with exploitation.

3.6 Positional Debiasing and Final Prediction

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* lane labels (LEFT $\leftrightarrow$ RIGHT) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$. The final prediction is $p_{\text{spare}} = \sigma(\delta_{\text{opt}}/T_{\text{dec}})$ where $\delta_{\text{opt}} = \bar{\delta} + \delta^*$ and $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; Table 1 lists all hyperparameters (validation details in Appendix D).

Algorithm 1 SWA-MPPI Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; threshold $\tau^{(c)}$; hyperparameters $K, \sigma, \lambda_{\text{coop}}, \alpha_{\text{KL}}, T_{\text{dec}}$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                    ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                    ▷ Debiasing
Compute consensus δ̄, rewards r_i, variance ν                 ▷ Eqs. 3–4
if ν ≥ τ^(c) then                                             ▷ Conflict detected
    for k = 1 to K do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)           ▷ Eqs. 6–8
    end for
    δ* = Σ_k w_k ε_k                                          ▷ Eq. 9
else  δ* ← 0
end if
return p_spare = σ((δ̄ + δ*)/T_dec)
```

Table 1: SWA-MPPI hyperparameters. PT parameters from Kahneman and Tversky [1979]; others validated on a held-out synthetic set (Appendix D).

Parameter	Symbol	Value	Source
Persona agents	N	4	Design
MPPI samples	K	128	Design
Perturbation σ	σ	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
KL penalty	α_{KL}	0.05	Validation
PT curvature	$\alpha{=}\beta$	0.88	Kahneman and Tversky [1979]
PT loss aversion	κ	2.25	Kahneman and Tversky [1979]
Decision temp.	T_{dec}	0.5	Validation
Trigger rate	ρ_{target}	0.35	Design
T_{cat} (Species / Gender / Others)		4.0 / 3.5 / 1.5	Category

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 2). For each of 15 countries, we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix E). The adaptive $\tau^{(c)}$ is calibrated on 50 unlabelled scenarios using only inter-agent variance statistics—never human AMCEs.

4.2 Models

Six LLMs: **Qwen2.5-72B-Instruct** [Qwen Team, 2025] (Int8), **Llama-3.1-70B-Instruct** [Grattafiori et al., 2024] (FP8), **Qwen2.5-7B/32B-Instruct** and **Llama-3.1-8B-Instruct** (4-bit), **Mistral-Large-2407** [Mistral AI Team, 2024] (4-bit). All on a single H100 80GB via Unsloth [Unsloth AI, 2024], greedy decoding, seed 42.

Native-language evaluation. Each country's scenarios are presented *entirely in the native language*: scenario framing, character descriptions, lane labels, and closing questions are all translated into 12 languages (zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es, en). Only the *decision tokens* (`LEFT`/`RIGHT`) remain English—these are verified as single-token entries in each model's vocabulary (Appendix F). This

Figure placeholder: SWA-MPPI Pipeline Overview

Left: WVS Persona Construction → Middle: Batched Forward Pass → Right: MPPI Optimisation Loop

Figure 1: **SWA-MPPI pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** MPPI optimisation when inter-persona variance exceeds $\tau^{(c)}$.

Table 2: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

design ensures that the moral reasoning is grounded in the target language's conceptual framing while maintaining a consistent, unambiguous decision interface across all models and languages.

4.3 Baselines

On Llama-3.1-70B: (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] using culturally contrastive steering vectors (details in Appendix G). Reward-guided methods (ARGS, controlled decoding) are excluded as they require reward models unavailable for cross-cultural settings (Appendix G).

4.4 Metrics

JSD (primary, ↓): Jensen-Shannon Distance between model and human AMCE vectors. **Pearson** r (↑): correlation of 6D AMCE vectors. **MAE** (↓): mean absolute AMCE error. All aggregated as unweighted means across 15 countries with bootstrap 95% CIs (1000 resamples).

5 Results

5.1 Cross-Model Comparison

Table 3 summarises performance across six models. The two 70B-class models benefit most: Qwen2.5-72B achieves **37.6%** mean JSD improvement (Pearson r: $0.548 \rightarrow 0.696$) and Llama-3.1-70B achieves **34.7%** (r: $-0.026 \rightarrow 0.384$). Smaller models show consistent +18–23% gains.

Table 3: Cross-model comparison. Bootstrap 95% CIs; differences $>\pm 0.004$ significant at $p<0.05$.

Model	Size	Mean JSD ↓		Improv. (%)	Pearson r ↑	
		Van.	SWA		Van.	SWA
Qwen2.5-72B	72B	$.0859 \pm .005$	$.0536 \pm .003$	**+37.6**	0.548	**0.696**
Llama-3.1-70B	70B	$.0937 \pm .006$	$.0612 \pm .004$	+34.7	$-.026$	0.384
Llama-3.1-8B	8B	$.0533 \pm .004$	$.0436 \pm .003$	+22.6	$-.247$	0.271
Qwen2.5-7B	7B	$.0522 \pm .004$	$.0426 \pm .003$	+18.4	0.252	0.402
Mistral-Large	126B	$.0457 \pm .003$	$.0430 \pm .003$	+5.8	0.658	0.550
Qwen2.5-32B	32B	$.0632 \pm .004$	$.0633 \pm .004$	-0.2	0.625	0.550

Table 4: Baseline comparison on Llama-3.1-70B across 15 countries.

Method	Mean JSD ↓	Pearson r ↑	ΔJSD vs SWA
SWA-MPPI (ours)	$\mathbf{.0612 \pm .004}$	**0.384**	–
Country-Tailored (B1)	$.0680 \pm .004$	0.233	+11.1%
PRISM-Style (B3)	$.0699 \pm .005$	0.148	+12.5%
WVS Profile (B2)	$.0737 \pm .005$	0.155	+16.9%
Activation Steering	$.0877 \pm .006$	-0.208	+30.2%
Vanilla	$.0937 \pm .006$	-0.026	+34.7%

Mistral-Large (+5.8%) has high baseline alignment leaving less room for correction. Qwen2.5-32B (-0.2%) exhibits concentrated logit distributions (low inter-token entropy at the decision position) causing the fixed $\sigma = 0.3$ perturbations to land outside the model's effective logit range, yielding near-uniform importance weights ($K_{\text{eff}} \approx K$) and reducing MPPI to random noise. A natural remedy is adaptive σ selection: setting $\sigma \propto \text{std}(\delta_1, \ldots, \delta_N)$ would scale perturbations to each model's logit spread, ensuring meaningful importance-weight differentiation. We leave empirical validation of this heuristic for future work (Appendix L).

5.2 Baseline Comparison on Llama-3.1-70B

Table 4: SWA-MPPI achieves the lowest mean JSD (0.0612) and highest Pearson r (0.384), outperforming the best alternative (country-tailored prompting) by 10.0%. Activation steering performs *below* vanilla, suggesting contrastive steering vectors interfere with moral reasoning on this benchmark.

5.3 Per-Country Results: Qwen2.5-72B

Table 5: SWA-MPPI improves JSD in 13/15 countries. **The Anglosphere paradox:** the largest gains occur for Australia (+75.4%), Great Britain (+72.2%), and USA (+71.1%)—precisely the English-speaking populations one might expect to be best served. Their high vanilla JSD (≈0.10–0.11) reflects RLHF-induced biases (extreme animal protection, strong gender-neutral stances) that diverge from average human preferences.

Germany (ceiling case): vanilla JSD = 0.0603 with $r = 0.845$—already exceptionally aligned. SWA-MPPI holds steady (JSD unchanged), demonstrating its *self-limiting* property: low inter-persona disagreement means the trigger rarely fires. The Pearson r decrease (0.845→0.747) reflects sensitivity to small rank perturbations in 6D, not distributional degradation (Appendix J).

Saudi Arabia (-2.3%) and Brazil (+6.7%) have limited WVS Wave 7 coverage, constraining persona quality.

6 Ablation Study

We ablate six components on Qwen2.5-72B, USA (342 scenarios; Table 6). Note: the ablation uses the full scenario pool, yielding JSD = 0.0425 vs. 0.0308 in Table 5 due to different dataset splits (Appendix H).

Table 5: Per-country results for **Qwen2.5-72B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1066	**.0308**	0.400	0.822	−.076	+71.1
Germany	de	.0603	.0603	0.845	0.747	+.000	+0.0
China	zh	.0513	.0371	0.851	0.762	−.014	+27.8
Japan	ja	.0986	.0553	0.634	0.786	−.043	+43.9
Brazil	pt	.0754	.0703	0.349	0.529	−.005	+6.7
Saudi Arabia	ar	.0778	.0796	0.443	0.527	+.002	−2.3
Vietnam	vi	.1302	.0783	0.569	0.456	−.052	+39.9
France	fr	.0789	.0479	0.617	0.838	−.031	+39.4
India	hi	.0600	.0561	0.670	0.739	−.004	+6.5
South Korea	ko	.0720	.0513	0.355	0.645	−.021	+28.8
Great Britain	en	.1057	**.0294**	0.419	0.800	−.076	+72.2
Russia	ru	.0800	.0629	0.623	0.784	−.017	+21.4
Mexico	es	.0760	.0534	0.605	0.545	−.023	+29.8
Nigeria	en	.1103	.0649	0.407	0.615	−.046	+41.2
Australia	en	.1047	**.0258**	0.438	0.841	−.079	+75.4
Mean		**.0859**	**.0536**	0.548	0.696	−.032	**+37.6**

Table 6: Ablation on Qwen2.5-72B, USA (342 scenarios). Full system: JSD = 0.0425, r = 0.724.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-MPPI	**0.0425**	–	**0.724**	–	**9.77**
1	No-MPPI (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on MPPI	0.0422	−.000	0.729	+.005	9.74
3a	Utility: quadratic	0.0429	+.000	0.725	+.001	9.68
3b	Utility: linear	0.0444	+.002	0.706	−.018	10.12
4	Without utilitarian (N=3)	0.0610	+.019	0.441	−.283	11.09
5	No debiasing	0.0476	+.005	0.743	+.019	9.67

The results establish a clear hierarchy: (i) the **utilitarian persona** is most critical (+43.5% JSD without it; Pearson r collapses by 0.283), anchoring the ensemble along the numerosity dimension; (ii) **MPPI** adds meaningful refinement (+15.9% JSD); (iii) **positional debiasing** corrects substantial recency bias (+12.0% JSD; Species preference inflated by 7.7pp without it); (iv) **Prospect Theory** contributes through nonlinear loss-averse weighting (+4.5% JSD vs. linear). The adaptive trigger fires at 100% for USA—a genuine empirical property of high-disagreement scenarios, not a calibration failure. The MPPI *flip rate* is only 6.0%, indicating the framework *modulates confidence* rather than reversing decisions (Appendix I).

7 Discussion

Key findings. SWA-MPPI achieves three desirable properties simultaneously: (i) substantial alignment gains on well-calibrated 70B models without weight modification, (ii) graceful self-limiting when intervention is unnecessary (Germany), and (iii) a 4.6× latency overhead (730ms vs. 160ms per scenario on H100) negligible for batch inference. The MPPI loop operates on cached scalars and contributes negligibly to wall-clock time.

The Anglosphere paradox. The largest gains occur for English-speaking countries—precisely those best-represented in training data. This reveals that RLHF introduces specific moral biases (extreme animal protection, strong gender-neutral stances) that *actively worsen* alignment with even Western human preferences, and SWA-MPPI's WVS-derived personas successfully counterbalance these.

Cross-model failure patterns. Per-country results (Appendix K) reveal that failure modes are *model-specific*, not country-specific. On Llama-3.1-70B, the largest gains shift to Vietnam (+65.8%) and Mexico (+66.4%), while Brazil (−23.3%) and Saudi Arabia (−10.7%) regress—a different

pattern from Qwen2.5-72B, where Brazil improves (+6.7%) and Saudi Arabia is the only regression (-2.3%). Mistral-Large shows a striking dissociation: JSD improves modestly (+5.8%) but Pearson r *decreases* for 11/15 countries (mean $\Delta r = -0.108$), indicating that the MPPI correction improves distributional distance while perturbing the rank ordering of AMCE dimensions. Qwen2.5-32B degrades in 7/15 countries, confirming the logit-concentration failure mode described in Section 5. These patterns suggest that SWA-MPPI's effectiveness depends on the base model's logit calibration quality, and that a pre-flight diagnostic (e.g., checking logit entropy) could predict when to activate the framework.

Limitations. We evaluate 15 of 100+ MultiTP countries; extending to low-resource languages remains untested. The per-agent reward r_i centres corrections relative to the base model, encoding an assumption that persona shifts approximate human directions—validated at aggregate level but unverified per-dimension. The WVS grounding encodes surveyed averages at a specific historical moment (Wave 7, $\sim$2017–2022); temporal staleness and limited WVS coverage for some countries constrain applicability. Extended limitations are in Appendix L.

Broader impact and safeguards. SWA-MPPI could improve fairness and inclusivity by calibrating LLM moral behaviour to diverse cultural preferences without costly retraining—the "Anglosphere paradox" insight alone has implications for how alignment protocols are designed and evaluated. However, the framework also risks aligning to harmful majority preferences (e.g., discriminatory norms against minorities within a population) or reifying stereotypes embedded in WVS survey averages. We identify three safeguards for responsible deployment: (i) *governance oversight*: SWA-MPPI should not override safety-critical model guardrails, and deployment decisions should involve domain experts familiar with the target culture; (ii) *periodic persona updates*: WVS data reflects a specific historical moment (Wave 7, $\sim$2017–2022), and personas should be refreshed as newer waves become available to prevent temporal staleness; (iii) *robustness checks*: practitioners should audit per-dimension AMCE shifts to detect cases where persona-driven corrections entrench harmful norms (e.g., extreme gender biases in certain WVS profiles). The WVS grounding provides a degree of empirical accountability—persona descriptions reflect surveyed population averages rather than author preferences—but this does not eliminate the risk of culturally legitimised harm.

8 Conclusion

We presented SWA-MPPI, a training-free inference-time framework for cross-cultural moral alignment. By grounding persona construction in WVS data, exploiting inter-persona disagreement as a trigger signal, and applying Prospect-Theory-weighted MPPI in logit space, SWA-MPPI reduces JSD to human preferences by 34–38% on 70B models, outperforms all baselines, and gracefully refrains from correction when alignment is already high. The Anglosphere paradox illustrates how RLHF biases worsen alignment even for well-represented populations. Future directions include: adaptive σ selection based on per-model logit entropy for broader compatibility; extension to multi-option preference settings; integration with newer WVS waves to assess longitudinal drift in moral preferences; disentangling language from culture effects via multilingual evaluation within single countries; and developing safeguards against entrenching discriminatory population-level norms—for instance, by flagging dimensions where persona-driven corrections exceed predefined ethical bounds.

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A AMCE Estimation Details

Binary dimensions. For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [Jin et al., 2025].

Utilitarianism dimension. Utilitarianism scenarios vary the *number* of lives on each side rather than character identity. Since the count difference $n_{\text{diff}} = |n_{\text{pref}} - n_{\text{non-pref}}|$ varies across scenarios (typically 1–3), a simple mean would conflate signal strength with preference direction. We therefore fit an OLS regression $p_{\text{spare}} = a + b \cdot n_{\text{diff}}$ and evaluate at the mean observed n_{diff} in the data. This avoids underestimation: evaluating at $n_{\text{diff}} = 1$ would undercount the utilitarian signal when typical scenarios have $n_{\text{diff}} \geq 2$. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded as non-discriminative.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata as follows:

1. **Variable extraction:** For each respondent, we extract 8 value dimension scores from specific WVS items: gender egalitarianism (mean of Q58P, Q59P, Q60P), religious importance (Q6P), interpersonal trust (Q43P), moral permissiveness (mean of Q50, Q52P, Q54P), work centrality (Q5P), family importance (Q1P), personal autonomy (Q39P), meritocratic orientation (Q40P). All items use inverted scales where higher = more progressive/positive.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded.

3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.

4. **Natural-language conversion:** Dimension means are converted to descriptors using calibrated thresholds (e.g., religiosity ratio $>0.85 \rightarrow$ "deeply religious"; trust ratio $>0.55 \rightarrow$ "high interpersonal trust").

5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

Indirect modulation: Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*—e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-MPPI does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and MPPI selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C MPPI Theoretical Justification

C.1 Degenerate but Principled Stochastic Optimal Control

The scalar decision gap δ constitutes a one-dimensional *state*; "dynamics" are identity (no state transition); perturbations ϵ_k are "control inputs" at a single horizon step. This is a degenerate case of MPPI [Williams et al., 2018], but the information-theoretic derivation remains valid: $w_k \propto \exp(U_{\text{total}}/\beta)$ minimises KL divergence between the sample distribution and the Boltzmann distribution of utility, making the KL term in Eq. 8 a direct consequence of the free-energy objective—not a heuristic penalty.

C.2 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. MPPI performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. The ablation (Table 6) provides indirect evidence: replacing PT with linear utility (closer to uniform weighting, hence a simple mean shift) degrades JSD by 4.5%.

D Hyperparameter Validation

Non-PT hyperparameters (σ, λ_{coop}, α_{KL}, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures.

E Dataset Preprocessing and Calibration

E.1 Preprocessing Sensitivity

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. A formal sensitivity analysis (varying capping ratios, up-sampling rates) is recommended for future work.

E.2 Calibration Limitation

The 50-scenario $\tau^{(c)}$ calibration set is drawn from the evaluation distribution. We mitigate overfitting risk: (i) calibration observes only a single scalar (variance), unlikely to overfit; (ii) it targets a fixed trigger rate, not alignment metrics; (iii) the 100% trigger rate for USA suggests τ is not sensitively tuned. A fully separate calibration corpus would provide stronger guarantees.

F Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer LEFT or RIGHT in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"LEFT"` and `"RIGHT"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: LEFT=22397, RIGHT=11572; Llama-3.1: LEFT=31266, RIGHT=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

G Baseline Implementation Details

G.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-MPPI.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We acknowledge that a more exhaustive search—varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement—could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-MPPI is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

G.2 Omitted Baselines

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] require trained reward models. In the cross-cultural setting, this demands either a separate reward model per country or culturally conditioned models—neither exists for MultiTP. SWA-MPPI's constraint is operating without labelled preference data, making these methodologically incompatible.

H Detailed Ablation Analysis

JSD difference with Table 5: The ablation uses the full 342-scenario pool; per-country results use the evaluation split after calibration reservation, yielding different scenario compositions.

Always-on vs. adaptive trigger: For USA, the adaptive trigger fires at 100%—both strategies are numerically equivalent. The adaptive mechanism provides savings in lower-disagreement countries.

Utilitarian persona: Removing it collapses Utilitarianism MPR from 61.2% to 48.0% (human: 76.6%), confirming it anchors the numerosity dimension.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

I Trigger Mechanism and Latency

The adaptive τ targets 35% activation but achieves 100% for USA (all scenarios contested). The *flip rate* (MPPI reversals) is 6.0%, indicating confidence modulation rather than decision reversal.

Mean latency: 730ms (SWA-MPPI) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; MPPI operates on cached scalars and contributes negligibly.

J Germany Pearson r Analysis

Germany's JSD remains unchanged (0.0603) while Pearson r drops from 0.845 to 0.747. In 6D, Pearson r is sensitive to rank perturbations: a single dimension swap reduces r by $\sim$0.1 while JSD (depending on magnitudes, not ordering) stays constant. Rare MPPI corrections introduce minor rank perturbations without affecting distributional distance.

K Per-Country Results for All Models

Tables 7–11 present per-country results for all evaluated models beyond Qwen2.5-72B (Table 5 in the main text).

L Extended Limitations

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed.

Country coverage. 15 countries across major cultural clusters, but no low-resource languages (Amharic, Khmer, Yoruba). Cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting MPPI stability.

Table 7: Per-country results for **Llama-3.1-70B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1561	.1234	0.012	0.197	−.033	+20.9
Germany	de	.0709	.0391	−0.455	0.685	−.032	+44.9
China	zh	.0459	.0311	0.406	0.802	−.015	+32.3
Japan	ja	.0457	.0416	0.496	0.466	−.004	+9.0
Brazil	pt	.0532	.0656	−0.370	0.348	+.012	−23.3
Saudi Arabia	ar	.0539	.0596	0.043	−0.210	+.006	−10.7
Vietnam	vi	.1546	.0529	−0.519	−0.135	−.102	+65.8
France	fr	.0500	.0516	0.250	0.394	+.002	−3.0
India	hi	.0442	.0337	−0.315	0.715	−.010	+23.6
South Korea	ko	.0893	.0436	−0.272	0.551	−.046	+51.2
Great Britain	en	.1568	.0918	0.012	0.286	−.065	+41.5
Russia	ru	.0449	.0413	0.539	0.637	−.004	+7.9
Mexico	es	.1265	.0425	−0.371	0.307	−.084	+66.4
Nigeria	en	.1582	.0988	0.109	0.497	−.059	+37.5
Australia	en	.1551	.1010	0.039	0.224	−.054	+34.9
Mean		**.0937**	**.0612**	−0.026	0.384	−.033	**+34.7**

Table 8: Per-country results for **Qwen2.5-7B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0531	.0448	0.048	0.531	−.008	+15.5
Germany	de	.0530	.0522	0.207	0.212	−.001	+1.4
China	zh	.0512	.0455	0.401	0.397	−.006	+11.2
Japan	ja	.0527	.0540	0.490	0.449	+.001	−2.3
Brazil	pt	.0445	.0331	0.415	0.837	−.011	+25.8
Saudi Arabia	ar	.0539	.0443	0.234	0.265	−.010	+17.9
Vietnam	vi	.0547	.0472	0.259	0.263	−.008	+13.6
France	fr	.0529	.0442	0.259	0.315	−.009	+16.5
India	hi	.0570	.0524	0.466	0.183	−.005	+8.1
South Korea	ko	.0492	.0329	0.239	0.363	−.016	+33.2
Great Britain	en	.0520	.0302	0.140	0.483	−.022	+41.9
Russia	ru	.0506	.0452	0.373	0.491	−.005	+10.5
Mexico	es	.0479	.0384	0.038	0.360	−.010	+19.9
Nigeria	en	.0551	.0411	0.101	0.425	−.014	+25.3
Australia	en	.0550	.0431	0.170	0.452	−.012	+21.7
Mean		**.0522**	**.0426**	0.252	0.402	−.010	**+18.4**

WVS temporal staleness. Wave 7 (∼2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions cause MPPI perturbations to be off-manifold, yielding near-uniform importance weights. Adaptive σ based on per-model logit entropy could address this.

Table 9: Per-country results for **Llama-3.1-8B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0470	.0333	−0.323	0.498	−.014	+29.1
Germany	de	.0716	.0529	−0.101	0.382	−.019	+26.1
China	zh	.0561	.0457	−0.082	0.388	−.010	+18.5
Japan	ja	.0518	.0519	0.545	0.395	+.000	−0.2
Brazil	pt	.0732	.0628	−0.222	0.109	−.010	+14.2
Saudi Arabia	ar	.0551	.0513	−0.252	−0.230	−.004	+6.9
Vietnam	vi	.0505	.0423	−0.330	0.303	−.008	+16.2
France	fr	.0515	.0612	−0.610	−0.220	+.010	−18.8
India	hi	.0756	.0564	−0.471	0.347	−.019	+25.3
South Korea	ko	.0522	.0419	−0.352	−0.062	−.010	+19.7
Great Britain	en	.0375	.0255	−0.065	0.457	−.012	+32.0
Russia	ru	.0560	.0434	−0.230	0.593	−.013	+22.5
Mexico	es	.0439	.0383	−0.079	0.344	−.006	+12.7
Nigeria	en	.0474	.0410	−0.229	0.303	−.006	+13.5
Australia	en	.0505	.0363	−0.156	0.328	−.014	+28.1
Mean		**.0533**	**.0436**	−0.247	0.271	−.011	**+22.6**

Table 10: Per-country results for **Mistral-Large-2407** (4-bit). Note: despite modest JSD gains, Pearson r decreases for 11/15 countries.

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0414	.0211	0.841	0.834	−.020	+49.0
Germany	de	.0378	.0291	0.734	0.675	−.009	+23.0
China	zh	.0572	.0618	0.824	0.585	+.005	−8.1
Japan	ja	.0457	.0535	0.814	0.387	+.008	−17.1
Brazil	pt	.0519	.0625	0.359	0.510	+.011	−20.4
Saudi Arabia	ar	.0238	.0315	0.724	0.671	+.008	−32.4
Vietnam	vi	.0521	.0428	0.768	0.571	−.009	+17.8
France	fr	.0481	.0475	0.683	0.628	−.001	+1.2
India	hi	.0457	.0527	0.772	0.631	+.007	−15.4
South Korea	ko	.0332	.0320	0.546	0.355	−.001	+3.6
Great Britain	en	.0418	.0331	0.853	0.730	−.009	+20.7
Russia	ru	.0552	.0566	0.768	0.631	+.001	−2.6
Mexico	es	.0472	.0481	0.748	0.642	+.001	−1.9
Nigeria	en	.0401	.0301	0.128	0.131	−.010	+25.0
Australia	en	.0516	.0338	0.865	0.682	−.018	+34.5
Mean		**.0457**	**.0430**	0.658	0.550	−.003	**+5.8**

Table 11: Per-country results for **Qwen2.5-32B-Instruct** (4-bit). SWA-MPPI degrades in 7/15 countries due to concentrated logit distributions.

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0447	.0440	0.887	0.738	−.001	+1.5
Germany	de	.0587	.0534	0.893	0.803	−.005	+9.1
China	zh	.0507	.0574	0.827	0.818	+.007	−13.1
Japan	ja	.0485	.0592	0.846	0.805	+.011	−22.0
Brazil	pt	.0717	.0645	0.552	0.422	−.007	+9.7
Saudi Arabia	ar	.0883	.1015	0.704	0.625	+.013	−14.9
Vietnam	vi	.0900	.0780	0.634	0.556	−.012	+13.1
France	fr	.0730	.0681	0.647	0.652	−.005	+6.7
India	hi	.0737	.0840	0.458	0.583	+.010	−13.8
South Korea	ko	.0529	.0303	0.106	0.064	−.023	+43.0
Great Britain	en	.0470	.0345	0.889	0.732	−.013	+26.5
Russia	ru	.0700	.0780	0.721	0.711	+.008	−11.3
Mexico	es	.0775	.0933	0.632	0.381	+.016	−20.3
Nigeria	en	.0641	.0512	0.124	0.156	−.013	+20.1
Australia	en	.0541	.0425	0.883	0.781	−.012	+21.4
Mean		**.0632**	**.0633**	0.625	0.550	+.000	**−0.2**